00-Project-Overview

BSS Purple Team SOC — Project Overview

Project: Modern SaaS Security & Threat Monitoring
Owner: Emilio Burgohy — Burgohy Security Solutions
Start Date: 2026-05-11
Status: Active

Objective

Build a production-ready hybrid Security Operations Center that detects, responds to, and documents real attacks against a simulated enterprise environment. The project bridges on-premises Proxmox infrastructure with Microsoft Sentinel via Azure Arc to create a fully operational Purple Team lab.

Environment Summary

Component	Details
Attacker	TREX-ATTACK — Kali Linux — 10.0.20.162
Targets	WS01, WS02, WS03 — Windows 11 — bsslab.local
Domain Controller	DC01 — Windows Server 2025 — 10.0.20.50
SIEM	Microsoft Sentinel — sentinel-bss-soc workspace
Log Pipeline	Azure Arc + AMA + DCR-BSS-Workstations
Cloud	Azure — RG-BSS-Lab

Project Phases

Phase	Title	Status
Phase 1	Infrastructure & Endpoint Protection	✅ Complete
Phase 2	Visibility & Audit Logging	✅ Complete
Phase 3	Governance & Shadow IT	✅ Complete
Phase 4	Unified SOC & Detection Engineering	✅ Complete
Phase 5	Purple Team Exercises	🔄 In Progress

Marketable Products

- Endpoint Security Setup
- M365 Security Health Check
- Shadow IT Audit
- Managed Security Monitoring
- Purple Team Assessment Report

01-Infrastructure

Infrastructure

Proxmox Cluster

Node	IP	VMs Hosted
ms01-node1	10.0.20.21	DC01 (VM 100)
ms01-node2	10.0.20.22	WS01 (101), WS02 (102), WS03 (103)
msa2-node3	10.0.20.23	Metasploitable (104), Scanner (105), JuiceShop (106)

Workstations

Machine	IP	OS	Domain
WS01-BSS	10.0.20.51	Windows 11	bsslab.local
WS02-BSS	10.0.20.52	Windows 11	bsslab.local
WS03-BSS	10.0.20.53	Windows 11	bsslab.local
DC01	10.0.20.50	Windows Server 2025	bsslab.local

Attacker

Machine	IP	OS	Role
TREX-ATTACK	10.0.20.162	Kali Linux	Red Team

Azure Arc Status

- ms01-node1: Connected ✅
- ms01-node2: Connected ✅
- msa2-node3: Connected ✅
- WS01-BSS: Connected ✅
- WS02-BSS: Connected ✅
- WS03-BSS: Connected ✅
- bss-admin: Connected ✅
- bss-juice: Connected ✅

Log Pipeline

- Azure Monitor Agent (AMA): Installed on WS01, WS02, WS03
- Data Collection Rule: DCR-BSS-Workstations
- Log Analytics Workspace: sentinel-bss-soc
- Event ID collected: 5156 (Filtering Platform Connection)

02-Sentinel-Setup

Sentinel Setup

Workspace

- **Name:** sentinel-bss-soc
- **Region:** East US
- **Resource Group:** RG-BSS-Lab

Data Connectors

Connector	Status
Microsoft Defender for Endpoint	Connected ✅
Microsoft Defender for Cloud Apps	Connected ✅
Office 365	Connected ✅
Azure Arc	Connected ✅

Analytics Rules

Rule	Severity	Status
BSS Port Scan Detection	Medium	Active ✅

BSS Port Scan Detection Rule

- **Event ID:** 5156 (Windows Filtering Platform Connection)
- **Trigger:** More than 5 connections from same IP in 1 minute
- **Schedule:** Runs every 5 minutes
- **Lookback:** Last 5 minutes
- **Entity Mapping:** Host (Computer), IP (IpAddress)
- **Alert Grouping:** All alerts grouped into single incident

Secure Score

- **Note:** Secure Score reflects the live BSS M365 Business Premium tenant (emilio@burgohysecuritysolutions.com)

- **Current:** 79.04%
- **This is separate from the Proxmox lab environment**
- Lab environment monitored via Sentinel/Arc only

Key KQL Queries

Port Scan Detection

```
SecurityEvent
| where EventID == 5156
| where TimeGenerated > ago(15m)
| summarize Count = count() by Computer, IpAddress,
bin(TimeGenerated, 1m)
| where Count > 5
| order by TimeGenerated desc
```

Heartbeat Check

```
Heartbeat
| where TimeGenerated > ago(30m)
| project Computer, TimeGenerated
| order by TimeGenerated desc
```

Attack-Report-Port-Scan-Detection

Attack 01 — Port Scan Detection

Date: 2026-05-12
Phase: Purple Team Exercise
Tactic: Reconnaissance
MITRE ATT&CK: T1046 — Network Service Discovery

Objective

Simulate an attacker performing network reconnaissance against domain-joined workstations and confirm detection in Microsoft Sentinel.

Attack Details

Field	Value
Attacker	TREX-ATTACK (Kali Linux)
Attacker IP	10.0.20.162
Tool	Nmap 7.99
Targets	WS01, WS02, WS03
Target IPs	10.0.20.51, 10.0.20.52, 10.0.20.53

Command Used

sudo nmap -sS -Pn --disable-arp-ping
10.0.20.51 10.0.20.52 10.0.20.53

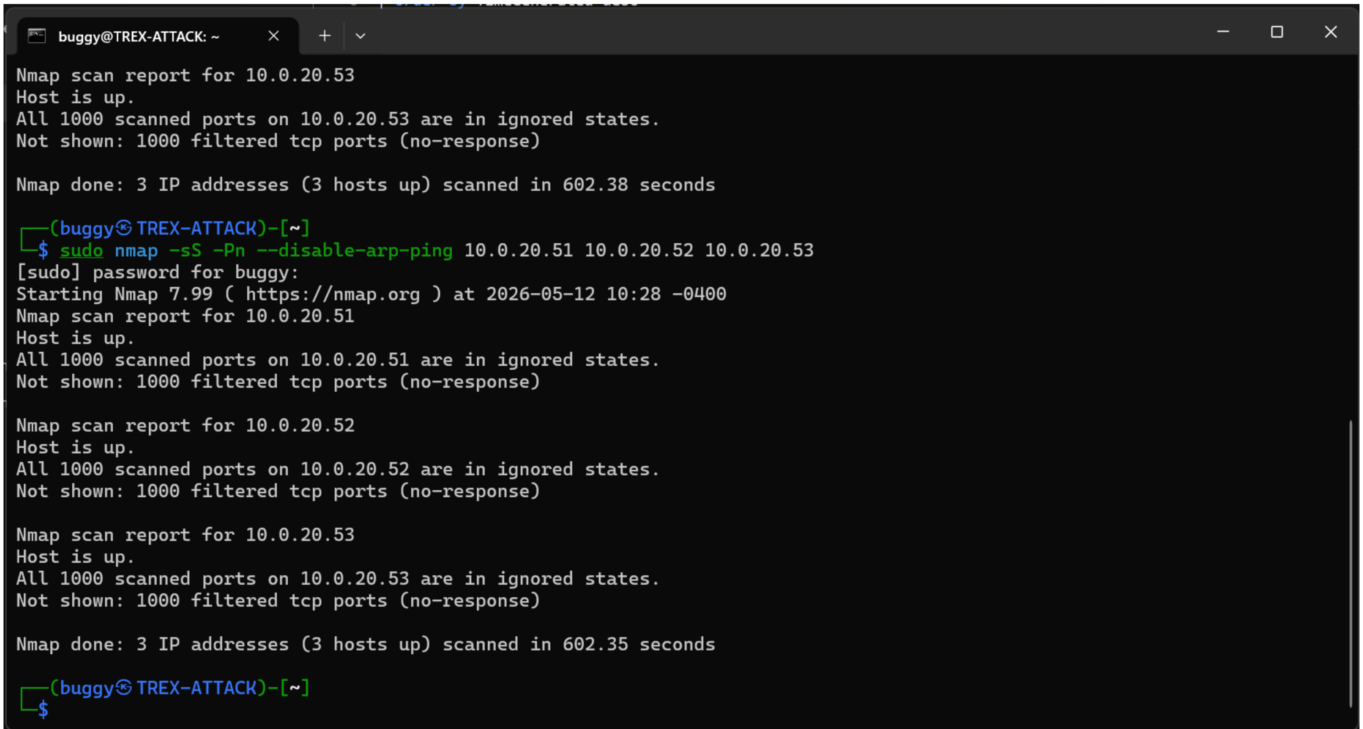
Detection

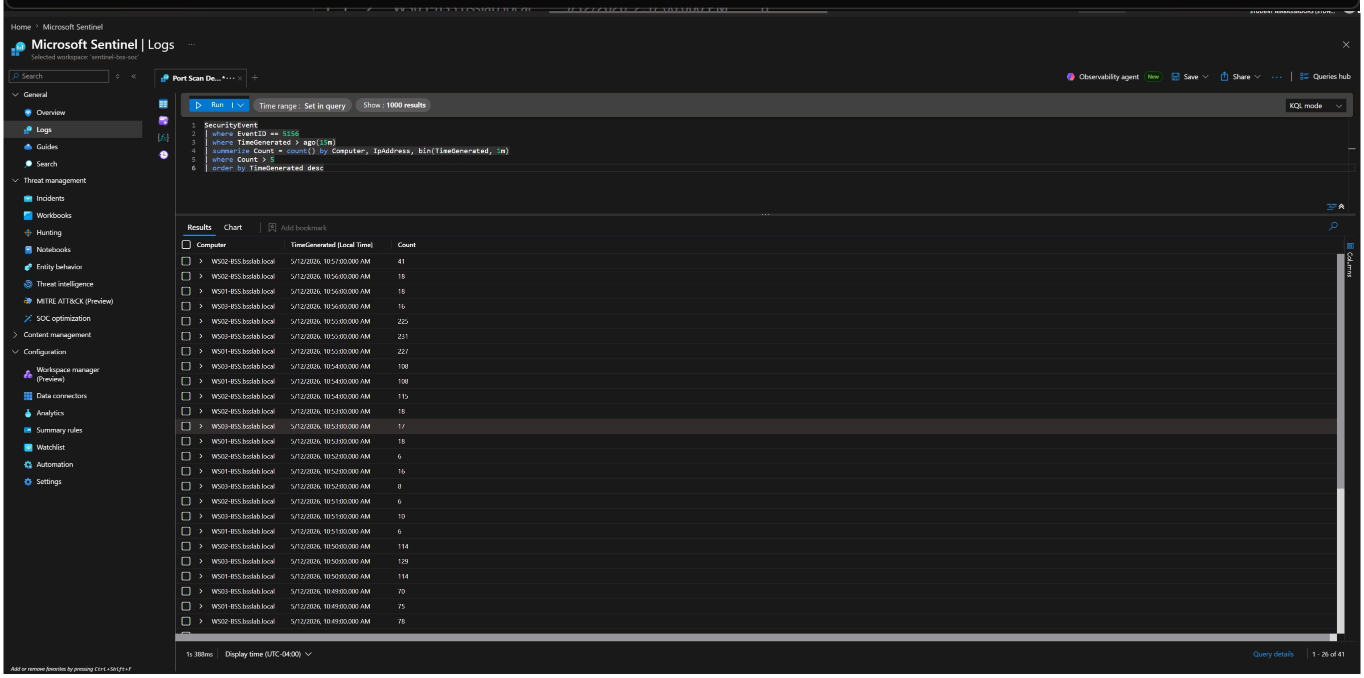
Field	Value
SIEM	Microsoft Sentinel
Event ID	5156
Analytics Rule	BSS Port Scan Detection
Severity	Medium
Peak Events	392 per minute on WS01

Result

DETECTED ✅

Screenshots





Incident-Report-BSS-Port-Scan-Detection

Incident Report — BSS Port Scan Detection

Date: 2026-05-11
Incident Numbers: 535 — 659
Total Incidents Generated: 124
Status: Closed
Classification: True Positive — Suspicious Activity

Summary

Microsoft Sentinel auto-generated 124 Medium severity incidents in response to an Nmap SYN scan performed from TREX-ATTACK against WS01, WS02, and WS03. Incidents fired every 5 minutes throughout the scan window due to the analytics rule schedule.

Root Cause

Nmap -sS full port scan from 10.0.20.162 against three domain-joined Windows 11 workstations triggered Windows Filtering Platform (Event ID 5156) logging at a rate exceeding the detection threshold of 5 connections per minute.

Timeline

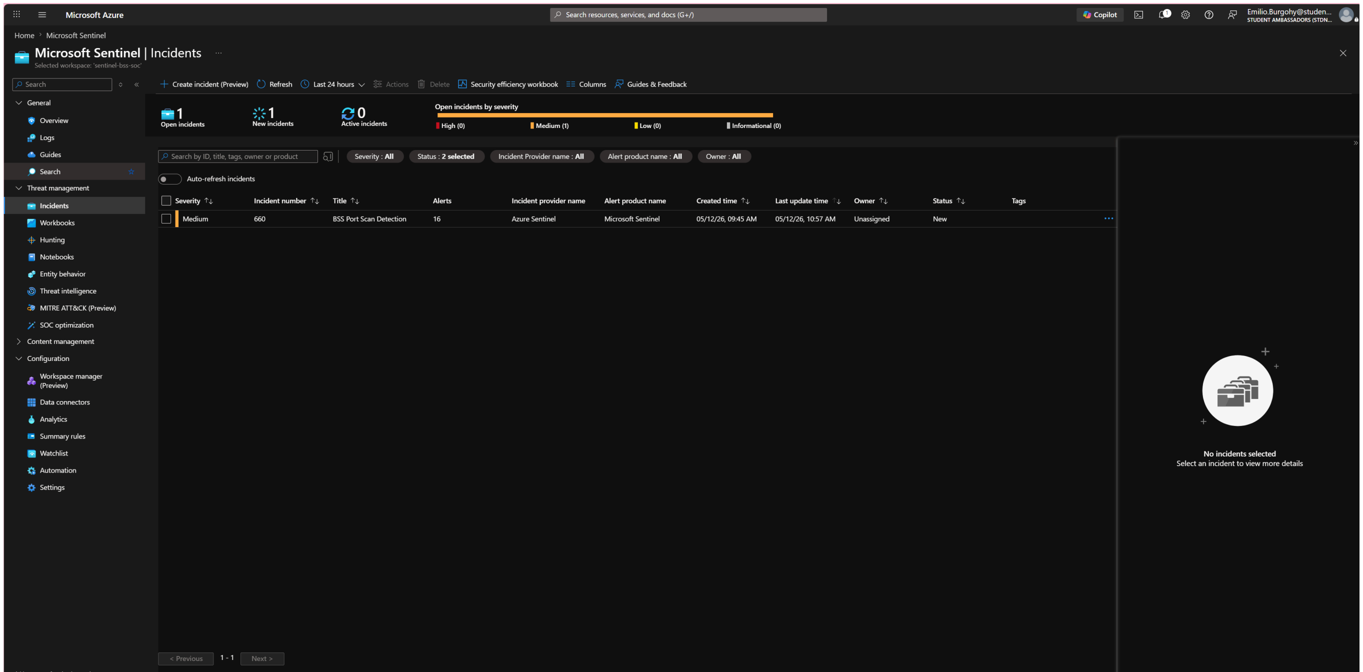
Time (UTC)	Event
05/11 03:39	Nmap scan initiated from TREX-ATTACK
05/11 03:41	First incident generated (#535)
05/11 03:51	Last scan incident generated
05/12 09:36	Final incident generated (#659)

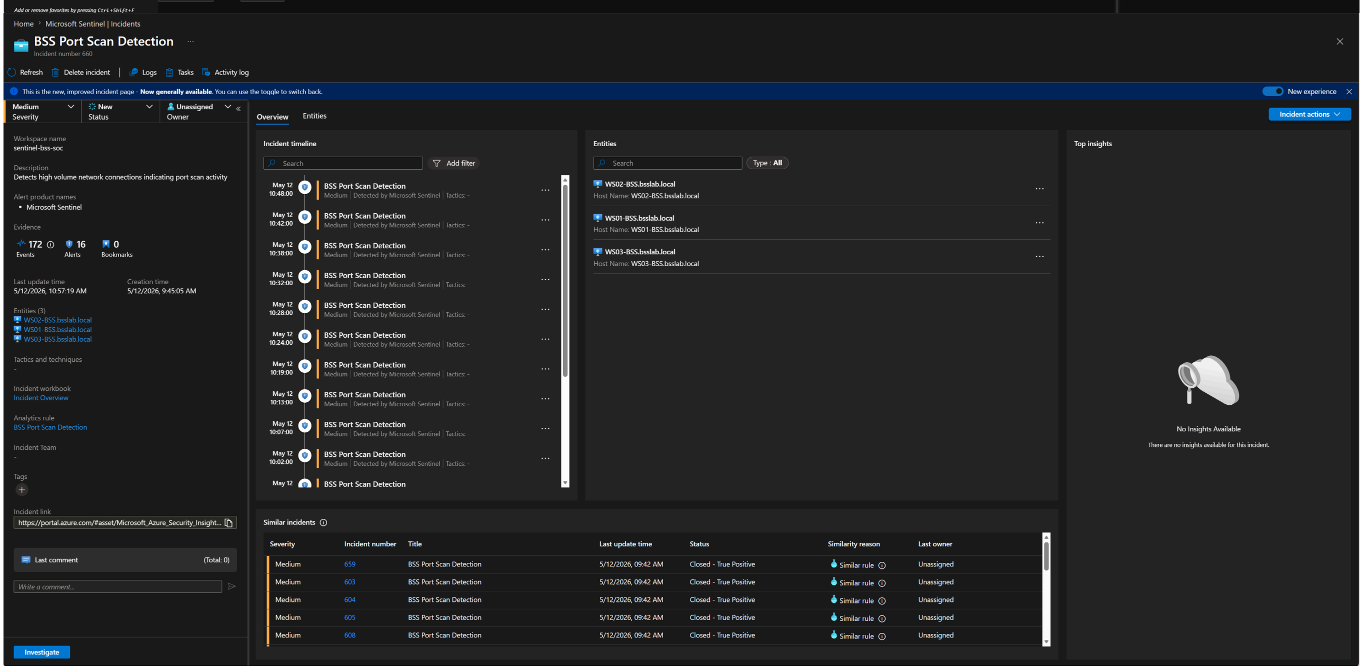
Lessons Learned

- Alert grouping was not configured — caused 124 individual incidents instead of one grouped incident
- Entity mapping was missing — investigation graph could not render attack path

- **Remediation applied:** Alert grouping set to single incident, entity mapping added for Host and IP

Screenshots



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Investigation
Undo
Redo
BSS Port Scan Detection
Incident
Medium
Severity
New
Status
Unassigned
Owner
5/12/2026, 10:57:19 AM
Last incident update time
Timeline
Info
Entities
Insights
Help